

Software Requirement Specification



Introduction

Project Title	MediConnect – Online Hospital Appointment Management System.
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Version	1.0

Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements for **MediConnect**, a web-based healthcare management system. The system aims to simplify healthcare services by allowing patients to book appointments, consult doctors, access medical records, and manage prescriptions online. It also provides doctors and administrators with tools to efficiently manage healthcare operations.

Project objectives

- Reduce appointment booking time by 70%
- Improve patient satisfaction Reduce manual workload for receptionists
- Increase online appointment adoption

Project Scope

Scope Breakdown	Details
Patient registration & Login	Customers can register or login in the app.
Doctor Search	Customers can search desired / available doctor.
Appointment booking	Customers can book desired appointment.
Appointment Cancellation	Customers can cancel their appointment if they wish.
Online payment	Multiple online payment options.
SMS/email notifications	Get notifications on important updates.

Intended Audience

- Business Analysts
- Project Managers
- Developers
- QA/Test Engineers
- UI/UX Designers
- Clients and Stakeholders

Overall Description

Product Perspective

MediConnect is a web-based healthcare management application that connects patients, doctors, and administrators through a secure platform.

The system integrates with:

- Payment Gateway
- Email/SMS Notification Service
- Hospital Database

User Classes

Patient

- Register and log in
- Search doctors
- Book appointments
- View prescriptions
- Access medical history

Doctor

- View appointment schedule
- Access patient records
- Create prescriptions
- Update consultation notes

Administrator

- Manage users
- Manage doctors
- Monitor appointments
- Generate reports

Operating Environment

- Windows
- macOS
- Android
- iOS

Supported Browsers:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari

Functional Requirements

User Registration

The system shall allow patients to create an account using:

- Full Name
- Email
- Phone Number
- Password

User Login

Users shall log in using:

- Email
- Password

The system shall validate credentials before granting access.

Doctor Search

Patients shall be able to search doctors by:

- Name
- Specialty
- Hospital
- Availability

Appointment Booking

Patients shall be able to:

- Select doctor
- Select available date
- Select time slot
- Confirm appointment

Appointment Management

Doctors shall be able to:

- View appointments
- Accept appointment
- Cancel appointment
- Mark consultation complete

Electronic Medical Records (EMR)

Doctors shall be able to:

- View patient history
- Add diagnosis
- Update treatment notes

Prescription Management

Doctors shall create electronic prescriptions including:

- Medicine Name
- Dosage
- Duration
- Instructions

Patients shall be able to download prescriptions.

FR-08 Billing & Payment

Patients shall:

- View consultation fees
- Make online payments
- Download invoices

Notifications

The system shall send notifications for:

- Appointment confirmation
- Appointment reminders
- Appointment cancellation
- Prescription availability

Admin Management

Administrators shall:

- Add/Edit/Delete doctors
- Manage patient accounts
- Monitor appointments
- Generate reports

Non-Functional Requirements

Performance

- System response time shall be less than 3 seconds.
- Support at least 1,000 concurrent users.

Security

- Passwords shall be encrypted.
- HTTPS shall be used for all communications.
- Role-based access control shall restrict unauthorized access.
- User sessions shall expire after inactivity.

Reliability

- Daily database backups shall be performed.
- System shall recover from failures without data loss.

Usability

- The interface shall be simple and user-friendly.
- Mobile-responsive design shall be supported.

Compatibility

The application shall support:

- Chrome
- Firefox
- Safari
- Edge